

HsiaoMan Liu

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Business Intelligence Engineer & Data Analyst

EDUCATION

University of Washington, Information School

Master of Science in Information Management

Specialization: Business Intelligence, Data Science, People Analytics and Program/Project Management | GPA: 3.8/4.0

Seattle, WA

June 2026

University of Washington, School of Public Health

Bachelor of Arts in Food Systems, Nutrition and Health

Minor: Data Science and Business Administration | GPA: 3.8/4.0 (Annual Dean's List 2022-2024)

Seattle, WA

June 2024

PROFESSIONAL EXPERIENCE

Graduate Researcher – People Analytics & Data Science

Sep 2025 – Present

University of Washington People Analytics Insights Lab

Seattle, WA

- Designed **ETL pipelines** for **1,000+** employee records using **Python (Pandas, NumPy)**, improving **data quality** by 40% and enabling accurate **predictive modeling** of performance trends and attrition risk.
- Conducted **time-series analysis** on employee performance, tenure, and engagement metrics to identify attrition patterns, delivering actionable insights that informed talent retention strategies and workforce planning.
- Led **business-oriented workforce analytics** research in a Web 3 company, applying **statistical tests** (ANOVA, regression, chi-squared) to identify engagement drivers and predictors of employee performance for **executive reporting**.

Operations Student Assistant

April 2023 – June 2025

University of Washington Department of Chemistry

Seattle, WA

- Drove **process improvement** reducing approval wait time by 50% (14 to 7 days) through **root cause analysis** on manual workflows, identifying bottlenecks, and leveraging **ServiceNow** ticketing system to eliminate process delays for **5,000+** users.
- Optimized financial reconciliation processes by analyzing **Workday ERP system data** using **advanced Excel** (VLOOKUP, INDEX/MATCH, Power Query), **automating data tracking** throughout the process which reduced data discrepancies by 35%.

Data and Business Intelligence Analyst

July 2022 – April 2023

Full-life Veterinary Hospital

Remote

- Built interactive **Tableau** dashboards visualizing client scheduling and visit data to identify peak cancellation windows and idle time slots, informing schedule restructuring that reduced no-shows by 32% and improved on-time appointment rates by 25%.
- Engineered **end-to-end ETL pipelines** and **star schema data models** in Snowflake using **SQL (window functions, CTEs)** across **10+** sources, reducing data processing time from 4 weeks to 14 days to support data-driven insight generation at scale.

PROJECT EXPERIENCE

Global Bike Inc. Business Intelligence Dashboard (2026)

- Developed **predictive discount optimization models** in **Advanced Excel Data Solver** on GBI transaction data, identifying 20% as the optimal discount rate to maximize profit through quantitative simulation and sensitivity analysis.
- Engineered data warehouse infrastructure in **Snowflake** by querying **SQL** to flatten transactional data enabling downstream analytics and dashboard development in **Power BI** using **DAX** and **Power Query** for sales and operations teams.
- Performed **data mining** on **10GB** of Amazon **customer sentiment data** using **MongoDB Atlas**, extracting aggregate product ratings and identifying quality trends across GBI product lines to support product decision-making.

Predictive Retention Analytics: Transforming Employee Experience in Roth's Retail (2026)

- Conducted **quantitative and qualitative analysis** on employee **survey data** identifying misalignment between employee self-reporting and manager evaluations, presenting actionable onboarding recommendations to the People Executive.
- Developed **predictive attrition models** using **logistic regression** achieving 72% accuracy and delivered targeted intervention strategies to Talent Management Leadership, including improved onboarding processes and manager effectiveness training.

Box Office Forecasting: Predicting Movie Revenue (2025)

- Engineered **predictive machine learning models** using **Python (scikit-learn)** on IMDb dataset with over 1000 records, comparing kNN, Random Forest, Decision Trees, and Logistic Regression; used R-squared and MSE to validate results.
- Performed **feature engineering** (categorical encoding, standardization, interaction terms) and visualized model insights using **Seaborn/Matplotlib**, identifying budget, genre, and release season as top revenue drivers.

SKILLS

- Programming Languages:** Python (Pandas, Numpy, Scikit-Learn, Pyspark, Pytorch), SQL, R
- Data Analysis & BI Visualization:** Tableau, Power BI, Streamlit, Looker Studio, Advanced Excel, Google Analytics, A/B Testing
- Database Management & Querying:** Snowflake, AWS (S3, Redshift), GCP (BigQuery), MySQL
- Data Engineering:** Git/GitHub, Airflow, Docker, Data Modeling (relational database, star schema), data warehousing
- Statistics:** Statistical Modeling, Statistical Testing, Machine Learning (Regression, Classification, Clustering)
- HRIS & Workforce Analytics:** Workday, ADP, UKG, ServiceNow